

Algebra PhD Exam, May 2007

Answer seven problems. You may quote results (within reason) as long as you state them clearly.

1. Give the definition of a normal algebraic field extension. Let E be a normal algebraic extension of the field F and $g(x) \in F[X]$ an irreducible polynomial. Prove that the irreducible factors of $g(x)$ in $E[X]$ all have the same degree.
2. Determine the Galois group of the polynomial $x^5 - 2 \in \mathbb{Q}[X]$.
3. Define the terms natural transformation and natural isomorphism between functors F and G from a category C to a category D . Show, giving full details, that $M \mapsto R \otimes_R M$ defines a functor from the category of left R -modules to itself, which is naturally isomorphic with the identity functor.

4. Let R be a ring with 1 and

$$0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0$$

be an exact sequence of left unital R -modules. Show that for any left R -module M , the sequence of induced maps

$$0 \longrightarrow \operatorname{Hom}_R(C, M) \longrightarrow \operatorname{Hom}_R(B, M) \longrightarrow \operatorname{Hom}_R(A, M)$$

is an exact sequence of abelian groups.

5. Prove that the following conditions on a ring R with 1 are equivalent:
 - (a) Every unital left R -module is projective.
 - (b) Every short exact sequence of left R -modules splits.
 - (c) Every unital left R -module is injective.
6. Let R be a commutative ring with 1 and I the intersection of all prime ideals of R . Prove that for each $a \in I$ there exists a natural number n such that $a^n = 0$.
7. Prove that the ring R of $n \times n$ matrices over a division ring is simple. Exhibit a minimal left ideal of R .
8. Define a primitive ring and state and prove the Jacobson Density Theorem.
9. Let R be a Noetherian commutative ring with 1. Prove that $R[X]$ is Noetherian.
10. Give examples of the following. Explain your examples, stating clearly any results you use, but you do not need to prove all details.
 - (a) A Noetherian ring which is not Artinian.
 - (b) A projective module which is not free.
 - (c) A Dedekind domain which is not a principal ideal domain.